

PALETTEDIFF: NATIVE-RESOLUTION PIXEL-ART SPRITE GENERATION WITH PALETTE-PROJECTED DIFFUSION

Anonymous authors

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ABSTRACT

Game sprites of 12–24 px are among the smallest images people draw on purpose, and their medium is defined by a handful of flat colours and hard alpha edges. Despite rapid progress in text-to-image generation, existing systems render such sprites as shrunken illustrations with soft shading and around a hundred colours. We identify two root causes: the training targets of sprite corpora are dominated by downsampled artwork rather than native pixel art, and continuous diffusion has no mechanism to express the few-colour structure of the medium. To address these challenges, we propose PaletteDiff, a framework for native-resolution sprite generation. First, we introduce *Native-Resolution Target Rebuilding*, which removes duplicate leakage, forbids aggressive downscaling and adds 15k permissively licensed sprites, together with a *native evaluation protocol* that scores against real low-resolution pixel art only. Second, we propose *Palette-Projected Sampling*, a training-free decoding rule that projects the predicted clean image onto a per-sprite k -colour palette during the late denoising steps, so that the remaining steps repair the quantisation seams. Third, combined with SDEdit-style re-noising, the same rule yields a *Palette Refinement* module for sprites from generators we did not train. PaletteDiff lowers FD-DINOv2 at 16 px from 167.2 to 31.8 ± 0.5 ; a per-sprite palette is the decisive ingredient, and imposing it inside the sampler improves on the same quantisation applied afterwards by a further 18%. Under the native protocol, PaletteDiff has the lowest FD among gpt-image-2, FLUX.2 and SDXL pipelines given the same palette post-process; Palette Refinement improves all three external systems by 36–41% while preserving 80% of their silhouettes.



Figure 1: 16 px RGBA sprites, nearest-neighbour magnified. **Row 1:** real sprites drawn at this size (median six opaque colours). **Row 2:** what a conventional sprite corpus feeds into its 16 px bucket — 25–64 px artwork box-downsampled (median 49 colours). **Row 3:** gpt-image-2 rendered at 1024 px and reduced. **Row 4:** PaletteDiff (4 colours), which reproduces the flat colour regions of native pixel art. Colour counts are per-row medians.

1 INTRODUCTION

Game sprites are among the smallest images anyone draws on purpose: an item, a character or an effect on a 12–24 px RGBA canvas, built from hard edges, a handful of flat colour regions and a transparent background, and a staple of 2D games and procedural content. At these sizes there is no texture in which to hide a mistake: one wrong pixel changes the silhouette, and one blended colour reads as blur. Recent text-to-image models (Podell et al., 2024; Black Forest Labs, 2026) have demonstrated remarkable capability at high resolution, and the natural recipe is to prompt such a model and shrink its output. The result is recognisable but it is not a sprite: it keeps the soft shading and the hundred-odd colours of an illustration (Figure 1, row 3).

We train a dedicated generator for this medium and identify two primary bottlenecks that hinder tiny-sprite generation. **1) Contaminated targets.** A sprite corpus assembled in the standard way — cutting sheets into connected components, bucketing by size and downscaling to fill each bucket — contains little native pixel art: in ours only 7.1% of the images are natively 16 px or smaller, so 92.9% of what the 16 px bucket sees is larger art box-downscaled into it, two thirds of it from 25–64 px. Exact duplicates additionally leave a pixel-identical twin in training for 16.2% of the held-out set. A model fitted to these targets learns soft, fifty-colour blobs, and an evaluation reference drawn from the same pool rewards exactly that. **2) Palette mismatch.** Real 16 px sprites use a median of six opaque colours and 46% of their adjacent opaque pixel pairs are exactly equal. A continuous diffusion model in RGB has no mechanism that snaps a region to a single value: every sampling rule we tested emits around fifty colours and reaches only 8% equal neighbours, spending the difference on few-level dither inside regions that should be flat.

To address these challenges, we propose PaletteDiff (Figure 2). *Native-Resolution Target Rebuilding* removes duplicate leakage, forbids feeding a sprite to a bucket more than $1.5\times$ below its native size, and adds 14,999 permissively licensed sprites, together with a *native protocol* that evaluates against real sprites drawn at the target size. *Palette-Projected Sampling* (PPS) then projects the predicted clean image onto a per-sprite k -colour palette during the late denoising steps, so the remaining steps repair the quantisation seams; it is training-free and transfers to a transformer backbone. Combined with SDEdit-style re-noising (Meng et al., 2022), the same rule is a *Palette Refinement* module for external generators.

At 16 px the rebuilt targets move FD-DINOv2 from 167.2 to 117.9, and PPS brings it to 31.8 ± 0.5 . Among practitioner pipelines used zero-shot — gpt-image-2, FLUX.2 with a pixel-art LoRA and SDXL with Pixel Art XL — PaletteDiff has the lowest native FD even when those systems receive the same palette post-process, and Palette Refinement improves all three by 36–41% while preserving 80% of their silhouettes. Our main contributions are:

- **Native-Resolution Target Rebuilding and the native protocol.** We diagnose target contamination and duplicate leakage in sprite corpora, and provide a rebuilt corpus and an evaluation protocol grounded in real low-resolution pixel art.
- **Palette-Projected Sampling.** A training-free decoding rule that enforces the few-colour structure of the medium inside the sampler, transfers to a transformer backbone, and outperforms post-hoc quantisation, a corpus-wide palette and de-dithering filters.
- **Palette Refinement.** SDEdit-style re-noising combined with PPS, which transfers the decoding rule to generators we did not train, with an explicit dial between preserving the input and matching the medium.

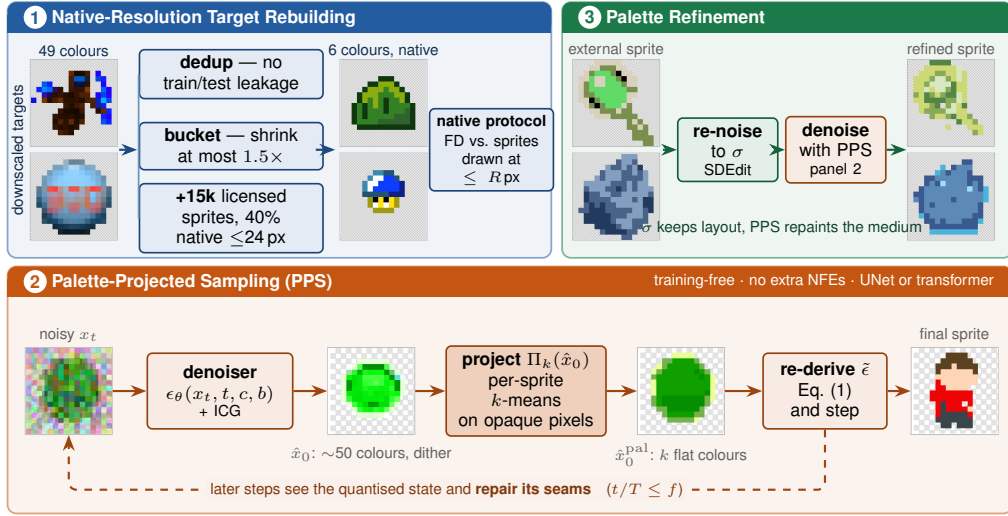


Figure 2: Overview of PaletteDiff. (1) Rebuild the targets and score only against sprites drawn at the canvas size. (2) Project $\hat{x}_0$ onto a per-sprite k -colour palette in the late steps; later steps repair the seams. (3) Re-noise an external sprite and denoise it with (2) to keep its layout while matching the medium.

2 RELATED WORK

Pixel art. Most work on pixel art converts an existing image rather than generating one: joint palette-and-pixel optimisation (Gerstner et al., 2012), contrast-aware downscaling (PixelOE; Yeh, 2024) and learned pixelization (Han et al., 2018; Wu et al., 2022). Those methods assume a finished picture; they never choose training targets for a generator, and they never invent a palette for a sprite that does not yet exist. The generative line is thin: character-sprite GANs (Coutinho & Chaimowicz, 2022), pose-conditioned sprite sheets (Hsieh et al., 2024), a DDPM conditioned on a given palette (Li, 2025), hours of score distillation from SDXL (Binnering & Sorkine-Hornung, 2024), and community LoRAs on large models (nerijs, 2023; Limbication, 2026). None of them ask whether a size-bucketed sprite corpus is dominated by downsampled artwork, and none estimate a per-sprite palette from the model’s own prediction in one sampling pass. PaletteDiff addresses both.

Guidance for diffusion models. Classifier-free guidance (Ho & Salimans, 2022) and its variants (Kynkäänniemi et al., 2024; Sadat et al., 2025a;c; Chung et al., 2025) change the reference that the conditional prediction is extrapolated away from; CADs instead anneals the condition (Sadat et al., 2024). Other rules use a weaker model (Karras et al., 2024), perturbed attention (Ahn et al., 2024; Hong et al., 2023; Hong, 2024), a degraded (Han et al., 2026) or a random condition (ICG; Sadat et al., 2025b). All of these leave the sample in a continuous colour space, so on this medium they still emit around fifty colours of dither. We adopt ICG, which needs no unconditional training, and compose it with a palette constraint that guidance alone does not provide.

Constraints inside the sampler. Projecting the predicted $\hat{x}_0$ is a standard device for inverse problems (Venkatakrisnan et al., 2013; Chung et al., 2023), and discrete-state diffusion imposes quantisation by changing the state space and the training objective (Austin et al., 2021; Gu et al., 2022). Neither line estimates a small palette on the fly from the current prediction and then lets a continuous denoiser repair the seams. PPS is that device; the backbone and its training stay unchanged.

Reference sets in generative evaluation. FID is sensitive to the feature extractor and to pre-processing (Heusel et al., 2017; Parmar et al., 2022; Stein et al., 2023), and duplicate leakage is a known corpus defect (Barz & Denzler, 2020; Webster et al., 2023; Torralba & Efros, 2011). Those analyses do not specify, for a size-bucketed medium, which real images belong in the reference. A mixed-size sprite reference is mostly downsampled illustration and reverses the ranking of methods. Following

Stein et al. (2023), we use DINOv2 features (Oquab et al., 2024); the native protocol additionally admits only sprites drawn at the canvas size.

3 METHOD

3.1 NATIVE-RESOLUTION TARGET REBUILDING

Diagnosis. Our starting corpus contains 37,489 unique sprites cut from OpenGameArt (OpenGameArt.org, 2025) sheets, together with the Kenney and Liberated Pixel Cup packs. Only 2,675 (7.1%) are natively at most 16 px and 12,516 at most 24 px; the remaining two thirds are 25–64 px artwork that a bucketed training pipeline box-downsamples into the low-resolution buckets. Downsampling averages neighbouring pixels, so these targets carry a median of 49 colours against six for sprites drawn at 16 px (Figure 1, rows 1–2). Only 4.6% of the corpus is integer-upscaled pixel art that could be restored by dividing by its block size. Hashing every sprite by its exact pixel content further reveals 4,888 duplicate copies; because a held-out set defined by file path cannot see that two paths hold the same image, 16.2% of the 3,000 held-out evaluation sprites have a pixel-identical twin in training.

Rebuilt targets. We (i) drop every duplicate copy and every copy of a held-out sprite; (ii) allow a sprite to feed a bucket only if it is at most $1.5\times$ larger than that bucket, so that a 64 px illustration can no longer become a 16 px target; and (iii) add 14,999 sprites cut from three permissively licensed OpenGameArt collections (CC-BY-4.0, CC-BY-SA-3.0, OGA-BY-3.0), of which 1,726 are natively at most 16 px and 6,028 at most 24 px, filtered for fonts, UI and near-identical animation frames and captioned with a vision–language model (13,987 usable captions, 99.6% distinct). Every added sprite keeps its source and licence in a manifest. The rebuilt corpus has 86,290 training rows and 65% more natively ≤ 16 px sprites than before (Appendix A).

Native protocol. The reference at resolution R is drawn only from real sprites whose native size is at most R , so no downscaling ever touches it: the natively $\leq R$ pool of the original corpus is split in half with a fixed seed into a reference and a disjoint floor set (at 16 px: 1,337 and 1,338 images, real-vs-real floor 4.47). The reference describes the target distribution and is drawn from the same pool as the training data, which it is not removed from; to measure generalisation without any overlap, we additionally evaluate on a *training-excluded native set* of held-out sprites that are natively ≤ 16 px and were removed from training (Section 4.5). We contrast the native reference with the *biased* reference obtained by sampling the whole corpus, of which 92.9% is downsampled at 16 px.

3.2 RESOLUTION-BUCKETED SPRITE DIFFUSION

We generate RGBA sprites on an $R \times R$ canvas, $R \in \{12, 16, 20, 24\}$, from a caption c . The denoiser $\epsilon_\theta(x_t, t, c, b)$ is a 72.5M-parameter UNet that operates directly on the four-channel image. Text enters through cross-attention to a frozen CLIP ViT-B/32 text encoder (Radford et al., 2021), and the resolution bucket $b \in \{12, 16, 20, 24, 32, 48, 64\}$ enters as a class embedding added to the timestep embedding. We train with the ϵ -objective (Ho et al., 2020) on a cosine schedule with $T = 1000$ and sample with 100 ancestral DDPM steps. The design is deliberately minimal, so that the gains we report can be attributed to the targets and the decoding rule; Section 4.3 shows that they transfer unchanged to a transformer backbone.

Guidance. All guidance rules combine the conditional prediction with a reference on the same state, $\tilde{e} = e_{\text{ref}} + w(\epsilon_\theta(x_t, t, c, b_R) - e_{\text{ref}})$. We use independent-condition guidance (ICG; Sadat et al., 2025b), in which the reference replaces the caption embedding by Gaussian noise with matched standard deviation, redrawn at every step. ICG needs no unconditional training, improves native FD over CFG 1.5 by 30.8, and composes with palette projection.

3.3 PALETTE-PROJECTED SAMPLING

A sprite is defined as much by how few colours it uses as by its shape. PPS enforces this property on the model’s own belief. At every step with $t/T \leq f$ we form the current clean-image estimate



Figure 3: Palette-Projected Sampling at 16 px, same prompts and seed; panel titles give the guidance weight and native FD of each row. Without projection (top), flat regions carry a few levels of dither. Projecting the predicted $\hat{x}_0$ onto four colours during the late steps (bottom) removes it, and the remaining denoising steps repair the seams, producing clean flat regions and crisp outlines.

$\hat{x}_0 = (x_t - \sqrt{1 - \bar{\alpha}_t} \tilde{e}) / \sqrt{\bar{\alpha}_t}$, cluster its *opaque* RGB pixels into k groups with a batched k -means (8 iterations, deterministic initialisation at evenly spaced luminance quantiles), replace every pixel by its centroid, leave transparent pixels and the alpha channel untouched, and map the result back to a noise prediction:

$$\hat{x}_0^{\text{pal}} = \Pi_k(\hat{x}_0), \quad \tilde{e} \leftarrow (x_t - \sqrt{\bar{\alpha}_t} \hat{x}_0^{\text{pal}}) / \sqrt{1 - \bar{\alpha}_t}, \quad (1)$$

before taking the scheduler step. The projection is a hard constraint on the model’s belief rather than a post-process: the steps that follow see the quantised state and repair the seams it creates (Figure 3), which is why PPS clearly outperforms quantising the finished image (Section 4.3). The palette is estimated per sprite, so each sprite keeps its own colours. PPS adds no network evaluations, since 8 k -means iterations over at most 576 pixels are negligible next to a UNet pass, and it requires no retraining. A pixel counts as opaque when its predicted alpha exceeds 0.5, the same threshold used to binarise alpha in the data. We use $f = 0.5$ throughout.

3.4 PALETTE REFINEMENT OF EXTERNAL SPRITES

Because PPS acts on the sampler state, it can be applied to an image the model did not generate. Following SDEdit (Meng et al., 2022), we re-noise a sprite produced by an external pipeline and reduced to the canvas to a fraction σ of the schedule, and denoise it with PaletteDiff under ICG and PPS. The parameter σ is an explicit dial: small values preserve the input, and $\sigma = 1$ discards it and recovers generation from noise, which we use only as a consistency check. This combines a large generator’s layout and subject with a sprite drawn in the grammar of the medium.

4 EXPERIMENTS

4.1 EXPERIMENTAL SETUP

Evaluation. For every configuration we draw one sample for each of 3,000 held-out captions, the recaptions of 3,000 held-out sprites. Our primary metric is the Fréchet distance of DINOv2-small CLS features (Oquab et al., 2024; Stein et al., 2023) (FD, lower is better) under the native protocol; images are composited on white and nearest-neighbour upsampled to 224 px. We also report CLIP ViT-B/32 image–text similarity ($100 \cos$) and retrieval R@1 of the own caption among 100. The seed-to-seed standard deviation of our headline configuration is 0.49 over three seeds; single-seed numbers therefore differ from the three-seed mean by a few tenths (e.g. 31.6 for seed 0). The palette size k , the start f and the guidance weight w were selected on the evaluation FD; at 16 px we verify that the selection of k transfers between disjoint halves of the captions (Section 4.3).

Baselines. We compare against three publicly available text-to-sprite pipelines: gpt-image-2 (OpenAI, 2026), FLUX.2-klein-4B (Black Forest Labs, 2026) with a pixel-art LoRA (Limbicnation,

Table 1: Comparison with external systems, 200 held-out captions (external generation is costly), native FD $\downarrow$. FD at $n=200$ is biased upwards relative to $n=3000$ (ours: 53.1 vs. 31.8), identically for every row. Every baseline receives the same k -colour post-process ($k = 4/8/6$ at 16/20/24 px) that PaletteDiff applies inside its sampler. Ours at 16 px is the mean of three seeds (50.8 / 52.9 / 55.4), 20 and 24 px are single seeds. The gray row, real held-out sprites of the original mixed-provenance corpus with the same post-process, is the ground truth of the conventional protocol and is shown for reference.

Method	16 px	20 px	24 px
SDXL + Pixel Art XL LoRA	244.2	526.3	527.2
FLUX.2-klein + pixel-art LoRA	150.1	473.9	472.4
gpt-image-2 + downscale	134.4	448.6	478.0
PaletteDiff (ours)	53.1 $\pm$ 2.3	152.5	192.2
Real sprites (mixed provenance)	77.0	165.5	150.7

2026), and SDXL (Podell et al., 2024) with Pixel Art XL (nerijs, 2023). Each generates at its native size and is reduced to the sprite canvas by the same cutout and downsampling code we use for real data. At 16 px, box downscaling is the strongest conversion we found: on gpt-image-2, measured against the corpus-sample reference, PixelOE (Yeh, 2024) and learned pixelization (Wu et al., 2022) give 146.3 and 155.2 against 82.2 for box downscaling (Appendix E). Every baseline additionally receives the same k -colour palette post-process that PPS applies inside our sampler. SD- π XL (Binniger & Sorkine-Hornung, 2024) requires about 8 GPU-hours per sprite and is therefore not evaluated at scale.

Implementation details. The main model is trained from scratch for 80k steps with AdamW at a constant learning rate of $5 \cdot 10^{-5}$, EMA 0.999 and 10% caption dropout. We use ICG with $w = 2$ and PPS with $k = 4$ at 16 px unless stated otherwise; Appendix C lists all hyperparameters.

4.2 COMPARISON WITH STATE-OF-THE-ART SYSTEMS

Table 1 and Figure 4 compare PaletteDiff, a specialist trained on sprites, with general-purpose generators used zero-shot, which is how practitioners obtain sprites today. PaletteDiff achieves the lowest native FD at every resolution, 2.5–4.6 $\times$ below every zero-shot baseline even though those baselines receive the same palette post-process; without it, gpt-image-2, FLUX.2 and SDXL score 284.4, 318.9 and 436.4 at 16 px. The post-process itself does not explain the gap: real mixed-provenance sprites with the same four-colour quantisation score 77.0, so quantisation alone does not reach our 53.1. That row is dominated by downscaled artwork (Section 3.1); against it, PaletteDiff is better at 16 and 20 px and behind at 24 px. Qualitatively, the external systems draw miniature paintings with soft gradients and anti-aliased outlines, while PaletteDiff produces the flat regions of sprites authored at this size.

4.3 ABLATION STUDY

Effect of each component. Table 2 shows that every component contributes. Rebuilding the targets alone improves FD from 167.2 to 117.9 without touching the sampler, and the new sprites account for a further 5.2 on top of the bucket policy. ICG brings it to 87.1, and PPS to 31.8 ± 0.5 , a step of 55.3. The components are complementary: PPS on the original targets reaches 42.9 and PPS with plain CFG reaches 58.4, while the full combination reaches 31.8.

In-sampler projection vs. post-processing. At ICG 1.5 (87.2 without a palette), quantising the finished samples to four colours already reaches 41.6, which identifies the per-sprite palette as the decisive property of the medium. Imposing the same palette inside the sampler reaches 34.1, a further 18%: the steps that follow the projection repair the seams that quantisation introduces, which a post-process cannot do.

Is it the per-sprite palette? Two controls replace Π_k at exactly the same steps. Projecting onto *one* four-colour palette shared by every sprite, learned by k -means over the reference set, gives



Figure 4: Qualitative comparison at 16 px on eight held-out captions. Rows: PaletteDiff; PaletteDiff without palette projection; gpt-image-2 generated at 1024 px, reduced and given the same four-colour post-process; FLUX.2-klein with a pixel-art LoRA, reduced *without* the post-process (Table 1 reports it with the post-process, 150.1); the real held-out sprite. The FLUX row is unquantised so the residual illustration look is visible; the table is the fair, post-processed comparison. Numbers in the row titles are single-seed native FD ($n = 200$).

Table 2: Ablation of the components of PaletteDiff, 16 px, 3,000 held-out captions: native FD $\downarrow$, and CLIP 100 cos / retrieval R@1 where measured (real held-out sprites: 27.66 / 22.6%). “Post-hoc” quantises the finished samples to the same palette size.

Targets	Guidance	Palette	Native FD $\downarrow$	CLIP / R@1
original	CFG 1.5	—	167.2	—
w/o new sprites	CFG 1.5	—	123.1	—
rebuilt	CFG 1.5	—	117.9	—
rebuilt	ICG 2	—	87.1	27.60 / 22.8%
original	ICG 2	PPS, $k=4$	42.9	—
rebuilt	CFG 1.5	PPS, $k=4$	58.4	—
rebuilt	ICG 1.5	post-hoc, $k=4$	41.6	—
rebuilt	ICG 1.5	PPS, $k=4$	34.1	—
rebuilt	ICG 2	PPS, $k=6$	36.9	27.38 / 22.0%
rebuilt	ICG 2	PPS, $k=4$	31.8 $\pm$ 0.49	27.07 / 19.2%

120.1; a 3×3 median filter, which removes the same dither without imposing a palette, gives 112.2. Both are far from PPS (31.6 with the same seed), confirming that the gain comes from each sprite’s own small palette rather than from fewer colours or smoothing alone.

Palette size and schedule. Sweeping k at 16 px gives 38.1 / 31.8 / 33.7 / 36.9 / 50.6 for $k = 3/4/5/6/8$: every setting improves over no projection (87.1) by a wide margin. The choice of k transfers across a validation split: selecting on one half of the captions picks $k = 4$, which is also best on the disjoint half. At $k = 6$ the samples reproduce the colour statistics of real sprites (six colours and 43% equal neighbours, vs. six and 46%) and are as retrievable from their captions as the real sprites (R@1 22.0% vs. 22.6%); $k = 4$ trades 2.8 points of retrieval for the best FD, and we recommend $k = 6$ when prompt fidelity matters most. PPS is also robust to its start point: single-seed runs give 31.4 and 31.6 at $f = 0.4$ and 0.5 (within seed noise), and running it for the entire schedule still gives 33.7.

Table 3: Other resolutions, native FD $\downarrow$ (ICG $w=2$). Ours: three seeds at 12–24 px.

Res.	w/o PPS	PaletteDiff (k)	Gain
12 px	155.7	143.2 ± 0.8 (8)	8%
16 px	87.1	31.8 ± 0.5 (4)	63%
20 px	117.0	96.5 ± 0.9 (8)	18%
24 px	186.8	123.6 ± 0.5 (6)	34%
32 px	97.1	74.5 (8)	23%

Table 4: Palette Refinement of external sprites, 16 px, native FD $\downarrow$ and silhouette IoU with the input at $\sigma=0.6$ (a fresh sample scores 0.46–0.53).

System	input	$\sigma=0.4$	$\sigma=0.6$	IoU
gpt-image-2	134.4	115.9	86.5	0.79
FLUX.2	150.1	116.1	88.2	0.80
SDXL	244.2	197.2	155.0	0.81



Figure 5: Palette Refinement on eight random captions. Rows: gpt-image-2 rendered large and reduced; the same sprite refined by PaletteDiff at $\sigma = 0.4$ and $\sigma = 0.6$; PaletteDiff generating the caption from noise; the held-out sprite. Refinement keeps the subject and layout of the input while converting its colours and edges to native pixel art.

Generality across backbones. We replace the UNet with a 69.7M-parameter transformer (PixArt-style adaptive norm; Chen et al. 2024), keeping corpus, schedule, sampler and evaluation unchanged, and compare the two at 20k steps with two seeds each. Under plain CFG the transformer trails by 36.5 FD (185.6 vs. 149.1); with ICG and PPS the transformer reaches 42.4 and the UNet 43.1 (seed spread at most 0.4), bringing two backbones that differ markedly under CFG to the same level.

Other resolutions. Table 3 shows that PaletteDiff improves native FD at every canvas size from 12 to 32 px. Absolute values are not comparable across rows: each resolution has its own native reference (half of the natively $\leq R$ px pool, e.g. 355 sprites at 12 px) and its own floor. The recipe is not specific to our target range: at 32 px PPS improves ICG from 97.1 to 74.5, and CFG scores 139.2.

4.4 PALETTE REFINEMENT OF EXTERNAL GENERATORS

Table 4 applies Palette Refinement to the three external pipelines. We measure retention as the IoU of the alpha mask with the input sprite, since DINOv2 similarity is saturated at this size. At $\sigma = 0.6$, native FD improves by 36–41% for all three systems while the mask IoU stays near 0.8, well above the 0.46–0.53 of a freshly generated sprite for the same caption. On gpt-image-2 the same σ takes native FD from 448.6 to 265.7 at 20 px and from 478.0 to 299.0 at 24 px (IoU 0.84 / 0.88). Sweeping σ is monotone (134.4 / 115.9 / 86.5 / 53.6 at $\sigma = 0/0.4/0.6/1$); $\sigma = 1$ recovers our own samples (53.6 vs. 53.1). At $\sigma = 0.6$, refinement retains most of the large generator’s caption retrieval (18.5% vs. 22.0%) while moving its style towards the medium (Figure 5).

Table 5: Per-image statistics at 16 px: median opaque colours and fraction of exactly equal adjacent opaque pixels.

	Colours	Flat pairs
Native reference (real)	6	0.46
Biased reference	34	0.20
gpt-image-2, reduced	93	0.01
CFG 1.5	52	0.08
ICG $w=2$	49	0.08
PaletteDiff, $k=6$	6	0.43
PaletteDiff, $k=4$	4	0.54

Table 6: Fraction of each set that a provenance probe (logistic regression on DINOv2 features), trained only on real sprites, calls natively drawn at 16 px.

Set	Native
Real held-out sprites (control)	94.9%
gpt-image-2, reduced	4.5%
FLUX.2 + LoRA, palette post-process	30.5%
gpt-image-2, palette post-process	33.0%
PaletteDiff w/o PPS	36.8%
PaletteDiff	67.5%

4.5 ANALYSIS

PPS reproduces the statistics of the medium. Table 5 shows why the decoding rule works. Every continuous sampler, as well as gpt-image-2, emits 49–93 colours with almost no exactly flat regions. PPS moves the samples from far outside the medium to inside it: at $k = 6$ PaletteDiff matches the real palette size exactly and reaches 0.43 flat pairs against 0.46 for real sprites.

A provenance probe. We fit a logistic regression on DINOv2 features to separate real sprites drawn at 16 px from 25–64 px artwork reduced to 16 px, using only real images and holding out the native pool used elsewhere. It reaches 93.0% held-out accuracy and calls 94.9% of unseen real sprites native. The probe shares its features with the FD and responds to palette size and edge hardness, so we read it as a complementary view of the same property rather than an independent judgement of quality. Table 6 shows that PPS nearly doubles the fraction of our samples classified as natively drawn (36.8% $\rightarrow$ 67.5%), while the external systems stay near one third even after quantisation. The same holds at 20 px (42.4% $\rightarrow$ 58.7%) and 24 px (28.1% $\rightarrow$ 42.7%), and Palette Refinement at $\sigma = 0.6$ raises gpt-image-2 from 33.0% to 53.0% and FLUX.2 from 30.5% to 58.5%.

Generalisation beyond the training pool. The native reference is drawn from the same pool as the training data. To rule out that the gains reflect proximity to training images, we evaluate on the 210 held-out sprites that are natively ≤ 16 px and were excluded from training, using their captions as prompts and the sprites themselves as ground truth; removing every prompt whose sprite has a near-duplicate in training leaves 124. The ordering is unchanged: PaletteDiff 61.1, ICG 76.2, CFG 1.5 97.7, against 34.4 for the real sprites themselves (all $n = 124$; Appendix G), so PPS closes 36% of the gap between ICG and real data on sprites the model has never seen. The same holds when prompts with near-duplicate twins are removed from the main evaluation (below).

The native protocol changes the conclusions. The biased reference is a population of blurred illustrations (34 colours, 0.20 flat pairs; Table 5), so it rewards samplers that produce them. A one-pass sampler that internalises guidance (Chen et al., 2025) is best under the biased reference and worst under the native one (Appendix B); PPS would be discarded by the biased one.

Robustness of the evaluation. Removing the 41.4% of prompts with a near-duplicate in training changes no ordering (PaletteDiff 31.8 $\rightarrow$ 29.0); an object-only protocol, Inception-v3 and longer training agree (Appendix G).

5 CONCLUSION

We proposed PaletteDiff for 12–24 px sprites at their native resolution. Native-Resolution Target Rebuilding and a native protocol fix contaminated targets; Palette-Projected Sampling enforces a per-sprite palette inside the sampler and, with SDEdit-style re-noising, refines external sprites. *Limitations.* A VLM judge prefers polished renders over real sprites (Appendix G), so a human study is the natural next step; PPS at $k = 4$ trades a little retrieval for fidelity ($k = 6$ matches real sprites).

REPRODUCIBILITY STATEMENT

Section 3.2 and Appendix C specify the architecture, training recipe and sampler, and Eq. (1) with Section 3.3 fully specifies Palette-Projected Sampling. Section 3.1 and Appendix A give every step of the data rebuild with per-step counts. Section 4.1 and Appendix C specify the evaluation protocol: prompt set, chunked seeding, reference sets, preprocessing and floors. Section 4.2 and Appendix E describe the conversion pipelines for the external systems, and Appendix B the cross-resolution samplers. We will release code for training, sampling, palette projection and evaluation, the captions, the per-sprite licence manifest of the new material and the held-out split.

ETHICS STATEMENT

The model generates small game sprites. It adds no capability beyond existing text-to-image systems, and its outputs (12–24 px) have little potential for targeted misuse. Its intended application raises the usual concerns of generative art tools, namely the displacement of pixel artists and the reproduction of styles present in the training corpus. The training data are community sprites from OpenGameArt and the Kenney and Liberated Pixel Cup packs. The 14,999 sprites added in this work carry a recorded source zip, member path and licence (CC-BY-4.0, CC-BY-SA-3.0, OGA-BY-3.0); the older material predates that bookkeeping, so a per-asset licence audit precedes any release of the full corpus. Captions were produced by a commercial vision–language model; external baselines were generated through commercial APIs under their terms of use. No human subjects were involved.

AI USE STATEMENT

Generative AI tools were used in this work in the following ways. (a) Inside the method and its data pipeline: vision–language models produced the training and evaluation captions (gemini-3-flash (Google DeepMind, 2025) for the original corpus, gemini-3.8-flash (Google DeepMind, 2026) for the material added here), labelled every evaluation sprite by content type and rated sprite craft for the ablations of Appendix G; commercial image generators served as external baselines (Section 4.2). Each of these uses is an experimental component, is described where it is used, and its outputs were spot-checked against manual inspection — the caption classifier was validated against image-level labels on 2,980 held-out sprites (82.5% agreement), and the content and craft labels were checked on random grids by the authors. (b) In the research process: an LLM-based coding assistant implemented and ran experiments, searched and verified related work, and drafted and edited parts of this manuscript. Every number reported here comes from a script in the released code and was regenerated from the saved samples; the authors reviewed all AI-assisted work and take responsibility for the final content, its claims and its errors.

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A THE REBUILT CORPUS, STEP BY STEP

Counts. The original corpus holds 37,489 unique sprites (after cutting sheets into connected components at native scale and dropping glyphs, UI and near-monochrome crops). Native sizes: 710 at most 12 px, 2,675 at most 16 px, 9,687 at most 20 px, 12,516 at most 24 px, 19,611 at most 32 px. Exact-pixel hashing finds 4,888 duplicate copies; 16.2% of the 3,000 held-out evaluation sprites and 17.0% of the 200 external-comparison sprites have a pixel-identical twin that the path-based hold-out leaves in training. Integer-upscaled pixel art, which could be restored by dividing by its block size, accounts for only 4.6% of the corpus.

New material. The CC-BY-4.0, CC-BY-SA-3.0 and OGA-BY-3.0 OpenGameArt collections (8.8 GB of 2D art) yield 37,226 cut sprites. Filtering by the same rules removes 8,184; cross-corpus pixel hashing removes a further 6,062 that duplicate the existing corpus; a filename filter removes 367 font, UI and text sheets; average-hash collapse of animation frames removes 3,456; a cap of 60 sprites per source sheet removes the rest, leaving 14,999 sprites from 1,207 sheets. Verification of this set: median palette 6 colours, 98.2% of visible pixels fully opaque, mean flat-neighbour fraction 0.434, native sizes 234 / 1,726 / 5,374 / 6,028 at most 12 / 16 / 20 / 24 px. Every sprite keeps its source zip, member path and licence in a manifest.

Captions. We caption the new sprites with gemini-3.8-flash on numbered 4×5 grids of nearest-upscaled sprites on a checkerboard, one caption per cell, 6–15 words, with an explicit “unclear abstract shape” escape. Of 14,999 sprites, 14,845 receive a caption (99.0%); dropping the escapes, degenerate strings and 11 rows that echoed the instruction leaves 13,987 usable captions, 99.6%

distinct, median 13 words. We selected the captioner by running the same grid through ten vision-language models: gemini-3.8-flash parsed all 20 cells in both trials at 22 s per call and was the most accurate, while a larger sibling truncated at cell 11 in both trials and several models mislabelled weapons and buildings.

Training-set composition. With duplicates removed and downscaling limited to $1.5\times$, the rebuilt corpus yields 86,290 training rows (12/16/20/24 px buckets: 7,357 / 11,733 / 15,087 / 17,199), against 64,990 without the new sprites and 180,533 under the original policy, which fed 48 px artwork to the 16 px bucket.

Near-duplicate collapse is harmful. Collapsing average-hash near-duplicates in the *old* corpus as well (37,489 $\rightarrow$ 26,504 groups) costs more than it gains: at 20k steps it scores 21.07 biased / 162.23 native against 20.31 / 151.85 for the same recipe without the collapse. We therefore drop only exact duplicates in the main corpus and apply the collapse only to the new material, where single sheets contributed up to 563 frames of one character.

B CROSS-RESOLUTION GUIDANCE: EQUATIONS AND FULL TABLES

The cross-resolution rules referenced in Section 4.5 use the guided prediction $\tilde{e} = e_{\text{ref}} + w(e_{\text{strong}} - e_{\text{ref}})$ with

$$e_{\text{ref}}^{\text{label}} = \epsilon_{\theta}(x_t, t, c, b_{\text{low}}), \quad e_{\text{ref}}^{\text{comp}} = \epsilon_{\theta_{\text{snap}}}(x_t, t, c, b_{\text{low}}), \quad b_{\text{low}} < b_R, \quad (2)$$

i.e. the same network (or an early EMA snapshot of it) queried under a lower resolution bucket, with $b_{\text{low}} = 12$ at 16 px and $b_{\text{low}} = 16$ at 20 and 24 px. The internalised variant follows guidance-free training (Chen et al., 2025): the network gains a scalar input $\beta \in (0, 1]$ and is trained with

$$\hat{e}_{\beta}(x_t, t, c, b) = \beta s_{\theta}(x_t, t, c, b, \beta) + (1 - \beta) \text{sg}[\epsilon_{\theta_{\text{snap}}}(x_t, t, c, b_{\text{low}}(b))], \quad (3)$$

so that setting $\beta = 1/w$ at inference reproduces the rule of Eq. (2) in a single forward pass. The internalised model is initialised from the base model and trained for 10k further steps (Table 7).

On the rebuilt model, scored against native pixel art, these rules give 114.9 (label), 117.0 (composed), 178.8 (internalised, $\beta = 1$) and 170.3 (internalised, $w = 1.5$), against 113.2 for CFG 1.0 and 87.1 for ICG; under the biased reference the internalised model is the best row of Table 9 (10.4). The reference is a lower-resolution belief, so extrapolating away from it adds contrast without reducing the palette, which the biased reference rewards and the native one does not.

C IMPLEMENTATION AND PROTOCOL DETAILS

Table 7: Training and sampling configuration.

Denoiser	diffusers UNet2DConditionModel, 4 input/output channels (RGBA), widths 128/256/512, 2 layers per block; down: CrossAttn, CrossAttn, plain; up: plain, CrossAttn, CrossAttn; 72.5M parameters
Text	frozen CLIP ViT-B/32 text encoder, 77 tokens, last hidden states (512-d) as cross-attention context
Resolution	class embedding over $\mathcal{B} = \{12, 16, 20, 24, 32, 48, 64\}$ added to the timestep embedding; never dropped
Diffusion	ϵ -prediction, $T = 1000$, cosine (squaredcos_cap_v2) schedule
Base training	from scratch, 80k steps, AdamW, constant $\text{lr } 5 \cdot 10^{-5}$, EMA 0.999, caption dropout 10%
Batches	one bucket per batch, bucket drawn proportionally to its row count; batch 320 / 256 / 224 / 192 / 128 / 64 / 40 for 12 / 16 / 20 / 24 / 32 / 48 / 64 px
Sampler	ICG $w=2$; PPS with $f=0.5$, $k=8/4/8/6/8$ at 12/16/20/24/32 px, 8 k -means iterations; opaque = $\hat{\alpha} > 0.5$
Snapshot	10k-step EMA of an independent 10k-step run with the same recipe (cross-resolution composed reference only)
Internalization	init from the base model, 10k steps, same optimizer/lr/EMA/batches; $\beta \sim \mathcal{U}(0.4, 1)$, $\beta=1$ w.p. 0.25 and for 12 px rows; reference rung 12 (for 16 px rows) and 16 (for 20 and 24 px rows); β via 256-d sinusoidal embedding of 1000 β and a zero-initialized linear map into the timestep features (32,768 parameters)
Sampling	100-step ancestral DDPM, EMA weights, one sample per prompt

Canvas preprocessing. A sprite larger than the target side R is reduced with a premultiplied-alpha box filter, preserving aspect ratio; a sprite at most $R/2$ is enlarged by an integer nearest-neighbour factor. It is then centred on a transparent $R \times R$ canvas, and alpha is thresholded at 128 (pixels below become fully transparent, with RGB cleared). Generated samples are stored with the same hard alpha. The same function produces the FD reference set, the held-out floor set and the “downscale” external baselines.

Evaluation prompts and training exclusion. All real sprites of the original corpus are split once with a fixed shuffle into a 3,000-sprite *biased* reference set and a disjoint held-out pool. The first 3,000 held-out sprites supply, through their recaptions, the 3,000 evaluation prompts. The biased reference and these 3,000 held-out sprites (5,928 paths) are removed from training, together with every exact duplicate of them.

Native reference. The sprites of the original corpus that are natively at most R px are split in half with a fixed seed into the native reference and a disjoint floor set (16 px: 1,337 and 1,338). This pool is not removed from training except where it overlaps the 5,928 excluded paths; the training-excluded native set of Section 4.5 consists of the floor-set sprites that are among the excluded paths (210 at 16 px).

FD. Each image is composited on white and resized to $R \times R$ (box filter; generated images are already $R \times R$). It is then nearest-neighbour upsampled to 224×224 , normalized with ImageNet statistics, and embedded by the CLS token of `facebook/dinov2-small`. FD is the Fréchet distance between Gaussian fits of the two feature sets (at 16 px: 3,000 samples vs. 1,337 native or 3,000 biased reference images; 200 samples for the external comparison). Samples are generated in chunks of 1,000 prompts, and chunk i uses seed $s + i$.

CLIP. The same saved samples are composited on mid-grey, nearest-neighbour upsampled to 224 px and scored with CLIP ViT-B/32 against the caption they were sampled from. We report $100 \cos$ and $R@1$, the fraction of samples whose own caption ranks first among itself and 99 random held-out captions (chance 1%).

D RECAPTIONING

Table 8: Caption statistics before and after recaptioning (43,342 unique training and evaluation images).

	BLIP (Li et al., 2022) (original)	gemini-3-flash (Google DeepMind, 2025) (recaptioned)
Distinct captions	23.5%	98.6%
Words per caption (median / mean)	8 / 7.6	12 / 12.1
Occurrences of the most frequent caption	7,669 (“a pixel art sprite”)	6 (“human”)
Share of data covered by the ten most frequent captions	$\approx 25\%$	$< 0.1\%$

Images were sent as numbered grids on a checkerboard (20 per request, 12 for retries). For each cell, the instruction asked for a caption of 6–15 words that names the object first and then its distinguishing colours, materials and parts, describes only what is visible, and avoids uninformative words (“pixel art”, “sprite”, “8-bit”, “icon”). Of the 43,342 unique images, 41,560 received a caption and 41,103 (94.8%) were usable. The remainder were declared an “unclear abstract shape” by the captioner and keep their BLIP (Li et al., 2022) caption. Across the three caption files (49,652 rows), 95% of rows were replaced. The total cost was about 18.7M tokens. The held-out evaluation prompts are the recaptions of the held-out sprites, so the evaluation prompts follow the training caption distribution.

E EXTERNAL SYSTEMS: PIPELINE AND QUALITATIVE COMPARISON

The gpt-image-2 (OpenAI, 2026) renders were requested at 1024 px through the vendor’s image-generation API with the prompt “<recaption>, pixel art sprite of a single object, plain flat white background, centred, no text”. The sprite is cut out by flood-filling, from the image border, all pixels within 18 (per channel, 8-bit) of the median border colour, and then cropped to its content. *Downscale* applies our canvas preprocessing (Appendix C). *PixelOE* pads the cut-out to a square on white, pixelizes the RGB in contrast mode with colour matching directly to $R \times R$, and takes alpha by majority vote of the cut-out mask in each cell. *Make Your Own Sprites* resizes the same square to $4R \times 4R$ (bicubic), applies the official network with 4 px cells, and reduces the output by 4 with nearest-neighbour sampling, with alpha as for PixelOE. We bypass the official script’s minimum input size of 128 px, because it would change the cell count for 12–24 px targets. The SDXL + Pixel

Art XL LoRA row uses SDXL base 1.0 with the fused LoRA (scale 1), Euler with 30 steps, CFG 7 and the same prompt and conversions.

Choice of conversion. We compared the three conversions on gpt-image-2 renders of the 200 captions against the corpus-sample reference, without palette post-process: box downscaling gives 82.2 at 16 px, PixelOE 146.3 and Make Your Own Sprites 155.2 (20 px: 317.6 / 282.5 / 336.7; 24 px: 428.5 / 358.0 / 431.2). Box downscaling is best at 16 px, where the main comparison is made, and we use it for every external system at every resolution.

F CAPTIONS OF THE QUALITATIVE FIGURES

Held-out prompt indices 3, 7, 12, 20, 33, 41, 55, 67, left to right in Figure 4:

1. Crou (*a caption the vision model could not resolve; kept to show the untouched prompt distribution*)
2. Alien soldier, blue with green armor, lunging forward while gripping a black rifle
3. Top-down character head with brown hair, black goggles, white antenna, and pale skin
4. Platypus like creature with brown body, yellow beak, and large eyes in swimming pose
5. Man: blocky person with short brown hair, red shirt, and dark blue pants
6. Stone brick wall with yellow rectangular bricks and a central red gem
7. Large fire, white-yellow core, intense orange glow, dark red jagged outlines
8. Glowing sun, bright yellow center with an orange border and a white cross

Figure 3 uses indices 11, 20, 33, 50, 67 of the same file.

G ADDITIONAL ANALYSES

Full 16 px table with both references. Table 9 lists every 16 px configuration under the native and the biased reference. The two columns rank samplers in opposite orders, which is the effect analysed in Section 4.5.

Table 9: 16 px, 3,000 held-out captions. *Native*: reference drawn only from sprites natively ≤ 16 px (floor 4.47). *Biased*: a random corpus sample (floor 3.45), 92.9% of which is downscaled artwork.

Targets	Sampler	Native FD ↓	Biased FD
original	CFG 1.5	167.2	11.1
w/o new sprites	CFG 1.5	123.1	14.8
rebuilt	CFG 1.0 / 1.5 / 2	113.2 / 117.9 / 131.1	19.0 / 13.6 / 12.2
rebuilt	cross-resolution label reference, $w=1.5$	114.9	14.2
rebuilt	composed reference, $w=1.5$	117.0	12.8
rebuilt	internalised (1 NFE), $w=1.5$	170.3	10.4
rebuilt	ICG $w=1.5$ / 2	87.2 / 87.1	16.2 / 20.7
rebuilt	ICG $w=1.5$ + post-hoc 4 colours	41.6	111.0
rebuilt	ICG $w=1.5$ + PPS $k=4$	34.1	111.0
rebuilt	ICG $w=2$ + PPS $k=6$	36.9	71.1
rebuilt	ICG $w=2$ + PPS $k=4$	31.8 ± 0.49	111.2

Palette size per resolution. Native FD with ICG $w=2$ for each palette size: at 12 px 143.2 ($k=8$), 143.3 ($k=6$), 145.2 ($k=10$), 150.7 ($k=5$), 156.5 ($k=4$); at 20 px 96.5 ($k=8$), 99.1 ($k=6$), 110.7 ($k=4$); at 24 px 123.6 ($k=6$), 128.4 ($k=4$), 130.1 ($k=8$); at 32 px 74.5 ($k=8$), 78.7 ($k=12$). Drawing k per sprite from the corpus histogram gives 39.9 at 16 px.

Native ground truth. A second evaluation set uses as ground truth the 210 held-out sprites that are natively ≤ 16 px and were excluded from training (124 after removing prompts whose sprite has a near-duplicate twin in training; Figure 6). On the 124 prompts the ordering is the same as in Table 2: CFG 1.5 97.7, ICG $w=2$ 76.2, PaletteDiff 61.1; the real sprites themselves score 34.4, which is the reference point of this set.



Figure 6: Twelve prompts from the training-excluded native set (sprites drawn at 16 px and removed from training). Rows: PaletteDiff; ICG only; FLUX.2-klein with a pixel-art LoRA, downsampled; the real sprite. Numbers in the row titles are native FD on all 210 prompts of this set, including those with near-duplicate twins in training; the text reports the 124-prompt subset without them.

Robustness of the evaluation. *Near-duplicate leakage:* removing the 41.4% of prompts whose sprite has a near-duplicate twin in training (8×8 average hash) changes every number by at most 3.0 FD and no ordering (PaletteDiff: 31.8 $\rightarrow$ 29.0). *Content:* restricting both reference and prompts to sprites a vision-language model labels as standalone objects gives 37.1 for PaletteDiff against 103.0 for ICG and 144.9 for CFG (see Content labels below). *Feature space:* Inception-v3 clean-FID preserves the ordering with smaller margins (below). *Training length:* 100k and 120k steps give 30.8 and 32.3, so the model is converged.

Inception features. Inception-v3 clean-FID (Parmar et al., 2022) and KID (Bińkowski et al., 2018) on the same saved samples and native reference preserve the ordering: PPS $k=4$ 23.79, $k=6$ 23.28, no projection 26.27, CFG 29.37, one corpus-wide palette 36.05, median filter 66.63 (real held-out 10.54). The separation is smaller than under DINOv2 because Inception resizes its input to 299 px, which blurs the hard edges and flat interiors that distinguish 16 px pixel art.

Content labels. A vision-language model labels every evaluation image (20 per numbered grid; classes object, effect, fragment, glyph, junk). Of the 3,000 held-out sprites, 73.2% are standalone objects, 15.7% fragments, 7.8% effects, 2.0% glyphs and 1.3% junk. The object-only protocol of Section 4.5 keeps 825 reference sprites and 2,181 prompts (PaletteDiff 37.1, ICG 103.0, CFG 144.9, original targets 201.3, real mixed-provenance sprites 161.3). Dropping the 8,818 sprites labelled fragment, glyph or junk from training improves a 20k-step run from 165.6 to 156.7 native FD; we keep the unfiltered corpus for simplicity.

Training length, capacity and data scale. The headline model is converged: 31.8, 30.8 and 32.3 native FD at 80k, 100k and 120k steps. A $2.2\times$ larger UNet (161.2M parameters) reaches 38.5 at 80k, so the 72.5M model is the better choice at this data scale. Adding the Kenney CC0 packs and the OpenGameArt CC-BY-3.0 bundle (118,378 training rows) improves a 20k-step run from 54.2 to 42.9 but gives 39.3 ± 0.3 at 80k against 31.8 ± 0.5 for the main corpus. Because both the capacity and the data-scale comparisons reorder between 20k and 80k steps, every headline number in this paper is measured at full training length. Replacing CLIP ViT-B/32 with ViT-L/14 as the text encoder gives 164.0 against 165.6 at 20k steps.

Quality conditioning. A vision-language model rates each sprite’s craft from 1 to 5, and we train a separate model with the rating prepended to the caption as a tag. Asking for the top tag at sampling time produces visibly more finished sprites (Figure 7). Because the tag concentrates the output on the best-rated 8% of the corpus, which is dominated by items and characters, distributional scores against the full native reference increase: the tag-trained model scores 45.6 with the top tag and 34.2

without it, against 31.8 for the main model. We therefore keep the main model unconditioned and offer the tag as a practitioner’s control.



Figure 7: Quality conditioning at 16px: same model, prompts and seed, sampled with (top) and without (bottom) the top craft tag.

Vision-language model as a judge. We also put 560 forced-choice pairs over 40 captions to a vision-language model, with real held-out sprites in the pool as a control. PaletteDiff ties with the real sprites (50%) under two different question wordings. The judge, however, also prefers the polished renders of FLUX.2 and gpt-image-2 over real sprites (72% and 70%), which indicates that a general-purpose vision model responds to polish rather than to whether an image was authored at sprite resolution; this is the property the native reference set is designed to capture, and it is why we do not use such a judge as a substitute for a human study.

Canvas occupancy. Real sprites drawn at 16px cover 26% of their canvas, while sprites that are resized to fill a bucket cover about 40%; PaletteDiff covers 34%. Preserving each sprite’s native margin during bucketing is a simple extension that we leave to future work.